CSA65 – GENERATIVE AI AND LARGE LANGUAGE MODELS

COMMON COURSE ASSIGNMENT – CO2, CO3 & CO4

Enterprise Knowledge Assistant Using Retrieval-Augmented Generation and AI Agents

Department	Computer Science and Engineering
Programme	B.E.– CSE-AI
Course Code & Course Name	CSA65 – Generative AI and Large Language Models
Academic Year / Batch	2026 – 2027
Faculty Name	**Dr. Malathi**
Assignment Title	Enterprise Knowledge Assistant Using Retrieval-Augmented Generation and AI Agents
Date of Issue	**02.09.2026**
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Maximum Marks	100
Course Outcome(s) – CO	CO2, CO3, CO4
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyze, L5 – Evaluate
SDG Mapping (SDG 1–17)	SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	Retrieval-augmented, agent-based knowledge assistants are being adopted across IT services, manufacturing, healthcare and banking to make large internal document repositories searchable, auditable and usable by every employee.

1. Problem Understanding and Formulation

KnowledgeDesk is an internal enterprise knowledge assistant for a 4,000-employee engineering services firm. It must answer natural-language questions over about 12,000 PDF, DOCX and scanned-image documents, cite the source document and page for factual statements, and abstain when evidence is insufficient. The design must integrate prompt engineering, retrieval-augmented generation (RAG), AI agents, multimodal processing and a secure intelligent application.

1.1 Challenges

- Prompting: produce consistent, grounded answers within an 8,192-token context window without fine-tuning.
- Retrieval: chunk, embed and index a growing corpus while achieving Recall@5 $\geq$ 0.85.

- Agents: handle multi-step requests such as comparing policy versions and raising a ticket.
- Multimodal processing: preserve information from scanned drawings, tables and screenshots.
- Performance: support 500 concurrent employees and keep 90% of responses within 3 seconds.
- Reliability/security: abstain below a confidence threshold, achieve ≥90% groundedness, enforce role-based access and prevent PII from being sent to external APIs.

1.2 Expected Outcomes

- Grounded, correctly cited answers.
- Recall@5 ≥ 0.85 on the evaluation set.
- Groundedness ≥ 90% for accepted answers.
- Correct tool use and stopping behaviour for multi-step requests.
- P90 end-to-end latency ≤ 3 seconds after optimisation.
- Auditable, role-aware and privacy-preserving retrieval.

1.3 Assumptions

Parameter	**Assumption**	**Justification**
Corpus	12,000 documents × 8 pages	Given scenario
Tokens/page	500	Given scenario
Chunking	512 tokens, 50 overlap	Given task; fits context
Embedding	768-D float32	Given constraint
Retrieval	Hybrid dense + BM25; top-20 candidates; rerank top-5	Balances recall and latency
Vector store	FAISS or ChromaDB	Open-source, practical for implementation
Application	FastAPI backend + Streamlit UI	Simple demonstrable architecture
Abstention	Evidence/confidence threshold + citation validation	Reduces hallucination

1.4 Constraints to Satisfy

- 8,192-token LLM context and one 24 GB GPU server.

- Rs.40,000/month hosted inference budget.
- ≤100 GB for vector index and document store.
- 500 concurrent employees and P90 ≤3 seconds.
- PII must not be sent to external APIs.
- Every factual statement requires document/page citation.
- AI disclosure, query/source logging and human escalation are required.

2. Application of Course Knowledge (CO2, CO3, CO4)

Part I – Prompt Engineering & LLM Behaviour (CO2, Unit II)

2.1 Four Prompt Variants

Common policy question used for all variants: "What is the process for applying for annual leave, and what should an employee do if the policy evidence is incomplete?"

A. Zero-shot

Answer the following KnowledgeDesk question using only the supplied context.
Question: What is the process for applying for annual leave, and what should an
employee do if the policy evidence is incomplete?
Context: [retrieved policy chunks]
If the context is insufficient, say: "I cannot answer reliably from the available evidence."
Cite the document name and page for every factual claim.

Expected output: a concise answer based only on the retrieved policy evidence, or an abstention if the evidence is insufficient.

B. Few-shot

You answer enterprise policy questions using only retrieved evidence.
Example:
Question: Where can an employee find the travel reimbursement limit?
Context: Travel_Policy.pdf, page 4: [evidence]
Answer: The limit is stated in the travel policy. [Travel_Policy.pdf, p.4]

Example:

Question: What is the leave approval process?

Context: No relevant evidence.

Answer: I cannot answer reliably from the available evidence.

Now answer:

Question: What is the annual-leave application process?

Context: [retrieved chunks]

Use document/page citations for factual claims.

Expected output: the examples teach consistent citation and abstention behaviour.

C. Chain-of-thought-style task prompt

Solve the policy question using only the retrieved evidence. Internally verify:

1) which evidence supports each claim,

2) whether any required step is missing,

3) whether the evidence is current and relevant.

Do not expose private reasoning.

Return only:

Answer: <grounded answer>

Evidence: <document and page citations>

Confidence: High / Medium / Low

If evidence is insufficient, abstain.

This uses internal verification without requesting private reasoning in the final answer.

D. Role-based system prompt

SYSTEM ROLE: You are KnowledgeDesk, a secure enterprise knowledge assistant.

GROUNDING: Use only authorised retrieved evidence in the current request.

CITATIONS: Every factual statement must include [Document Name, p.X].

ABSTENTION: If evidence is missing, conflicting, stale or below threshold, abstain.

SECURITY: Never reveal content outside the user's authorised department.

Never send PII to external services.

OUTPUT: {answer, citations[], confidence, abstained}

TASK: Answer the employee's question from the retrieved evidence.

Recommended production strategy: use the role-based system prompt as the permanent control layer and add a small number of few-shot examples for formatting. This gives stronger grounding and consistency than zero-shot alone without requiring fine-tuning.

2.2 Context Window Calculation

$400 + 600 + 512k + 600 \leq 8192$

$512k \leq 6592$

$k \leq 12.875$

Maximum integer k = 12 chunks.

With k = 12, the allocated context is 400 + 600 + 12×512 = 7,144 tokens. The remaining 1,048 tokens provide room around the required 600-token answer budget.

2.3 Monthly Cost

Input tokens/query = 400 + 600 + (12×512) = 7,144

Input cost/query = 7,144/1,000 × Rs.0.18 = Rs.1.28592

Output cost/query = 600/1,000 × Rs.0.55 = Rs.0.33

Total/query = Rs.1.61592

Monthly cost = 20,000 × Rs.1.61592 = Rs.32,318.40

Rs.32,318.40 is below the Rs.40,000 budget, leaving Rs.7,681.60 headroom. This is a worst-case calculation assuming every query consumes the full allocation.

2.4 Prompt Selection and Hallucination Controls

- Use a low temperature such as 0.1–0.2 for policy answers.
- Use low-variance decoding and avoid unnecessary sampling.
- Return a structured schema containing answer, citations, confidence and abstained.
- Abstain when retrieval confidence/evidence coverage is below the chosen threshold.
- Validate that every factual claim has a supporting document/page citation.
- Apply access-control filtering before evidence reaches the LLM.

Part II – Retrieval-Augmented Generation & AI Agents (CO3, Unit III)

2.5 Corpus and Vector-Index Calculation

Pages = 12,000 × 8 = 96,000

Total tokens = 96,000 × 500 = 48,000,000

Chunk step = 512 − 50 = 462

Chunks/document = ceil((4,000 − 512)/462) + 1

= ceil(3,488/462) + 1

= 8 + 1 = 9

Total chunks = 12,000 × 9 = 108,000

Bytes/vector = 768 × 4 = 3,072 bytes

Raw vector bytes = 108,000 × 3,072 = 331,776,000

≈ 0.332 GB decimal (≈0.309 GiB).

The raw vectors are far below the 100 GB storage limit. Metadata, OCR text, indexes and document files add overhead, so the final implementation should measure actual database size before submission.

2.6 Evaluation Set

ID	Representative question	Known source
Q1	How many days of annual leave are available?	HR Leave Policy
Q2	Who approves annual leave?	HR Leave Policy
Q3	What is the travel reimbursement limit?	Travel Policy
Q4	What is the quality inspection procedure?	Quality SOP
Q5	What is the product calibration interval?	Product Manual
Q6	What changed in the current leave policy?	Current + Previous Leave Policy
Q7	What are the document retention requirements?	Compliance Policy

Q8	How is a failed quality inspection recorded?	Quality SOP
Q9	Which department owns the equipment manual?	Product Manual
Q10	What is the escalation process for a compliance issue?	Compliance Policy
Q11	Which form is required for a travel claim?	Travel Policy
Q12	Where is the emergency shutdown procedure?	Safety/Equipment Manual

For an actual experiment, each question must be mapped to its gold source chunk/page. The assignment asks for measured Precision@5, Recall@5 and MRR. The following values are an illustrative template only and must be replaced with real execution results.

Configuration	**Retrieval**	**Precision@5**	**Recall@5**	**MRR**
A	512 tokens; top-5 dense	0.72*	0.83*	0.78*
B	512 tokens; top-10 hybrid dense+BM25	0.80*	0.92*	0.87*
C	384 tokens; top-20 hybrid + top-5 rerank	0.84*	0.96*	0.91*

*Illustrative values, not measured results. Do not submit them as experimental evidence.

Precision@5 = relevant retrieved results / 5. Recall@5 = relevant retrieved gold results / total gold relevant results. MRR is the mean reciprocal rank of the first relevant result.

2.7 AI Agent Design

- Vector retriever – retrieves authorised evidence.
- Document-metadata lookup – checks document version, date, owner and page metadata.
- Calculator – performs arithmetic/comparisons deterministically.

- Ticket-creation API – performs an approved external action after evidence is verified.
- Reasoning loop – bounded ReAct-style Plan/Act → Tool Result → Verify → Next Action.
- Stopping condition – all requested sub-tasks complete and evidence/action results are verified; otherwise abstain or escalate.
- Routing – plain RAG for single-step factual questions; agent for comparisons, calculations, multiple dependent tools or actions.

To protect latency, the agent is an exception path. Independent tool calls should run in parallel where possible, and the number of iterations should be capped.

Part III – Multimodal GenAI & Intelligent Applications (CO4, Unit IV)

2.8 Multimodal Approach Comparison

Approach	**Advantages**	**Limitations**	**Suitability**
OCR → text embedding	Simple; exact text searchable; easy citations	Weak visual/layout understanding; poor for drawings	Good baseline
Vision-language captioning → text embedding	Captures visual meaning; useful for screenshots/tables	May omit exact values/layout; adds inference cost	Good mixed-content option
Native multimodal embedding	Joint semantic representation of text and images	More complex; model/resource requirements	Best long-term visual retrieval

Recommendation: use a hybrid multimodal pipeline. OCR preserves exact searchable text and table values; vision-language processing captures visual meaning from drawings and screenshots. If a native multimodal embedding model fits the 24 GB GPU and latency budget, it can be added for visual retrieval while OCR remains available for exact citation/verification.

2.9 End-to-End Latency

Generation time for 250 tokens at 45 tokens/second = 250/45 = 5.556 seconds = 5,556 ms.

Pipeline	**Embedding**	**Vector search**	**Reranking**	**Generation**	**Total**	**≤3 sec?**

A – no reranking	40 ms	25 ms	0	5,556 ms	5,621 ms	No
B – top-20 rerank	40 ms	25 ms	2,400 ms	5,556 ms	8,021 ms	No
C – top-5 rerank	40 ms	25 ms	600 ms	5,556 ms	6,221 ms	No

The requirement cannot be met under the stated 45-token/second generation rate and 250-token answer length. Generation alone exceeds 3 seconds. Therefore the bottleneck is generation, not vector search.

With no reranking:
Maximum generation time = 3,000 − 40 − 25 = 2,935 ms
Maximum tokens at 45 tokens/s ≈ 2,935/1,000 × 45 ≈ 132 tokens.

A practical fix is to limit normal answers to roughly 120–130 tokens, use a faster model/runtime, or both. For example, at 120 tokens/second with top-5 reranking: 40 + 25 + 600 + (250/120×1000) ≈ 2,748 ms.

2.10 Intelligent Application Concepts

- FastAPI backend for query, feedback, health and controlled tool endpoints.
- Streamlit/web UI for a responsive employee chat interface.
- Inline document/page citations beside factual claims.
- Conversation memory using summarised, authorised context to control token growth.
- Feedback capture for helpful/not-helpful responses and escalation.
- Audit logs containing user role, query, retrieved sources, scores, citations and tool actions.
- Human escalation for low-confidence, sensitive or ambiguous requests.

3. Consolidated Solution / Design / Methodology

3.1 Prompting Design

Permanent system controls:

• Use only authorised retrieved evidence.

• Cite every factual statement as [Document, p.X].

• Abstain on insufficient/conflicting evidence.

• Do not expose private reasoning.

• Return answer + citations + confidence + abstained.

• Use temperature ≈ 0.1–0.2 for policy answers.

3.2 Retrieval Design

1. Collect PDF/DOCX/image documents and attach document ID, department, version and access-control metadata.
2. Extract digital text and OCR scanned pages; process tables and images separately.
3. Create 512-token chunks with 50-token overlap while preserving document/page/chunk metadata.
4. Generate 768-dimensional embeddings and store them in FAISS/ChromaDB.
5. Apply role-based filtering before evidence is passed to the LLM.
6. Use hybrid dense + BM25 retrieval for high recall; rerank only a small candidate set.
7. Validate evidence and citations before returning the answer.

3.3 Agent and Application Flow

```
Employee Query
   ↓
Authentication + Role Check
   ↓
Query Router
  ┌── Simple query → Hybrid RAG → Rerank → LLM → Citation Check → Answer
  └── Multi-step → Agent
          ├─ Retriever
          ├─ Metadata Lookup
          ├─ Calculator
          └─ Ticket API
             ↓
        Verify Evidence/Action
             ↓
       Answer / Abstain / Human
```

The architecture separates authentication, retrieval, reasoning and presentation. Access control is applied before evidence reaches the model.

4. Use of Modern Tools

Component	Tool	Purpose
Programming	Python	Ingestion, retrieval and evaluation
RAG framework	LangChain	Chunking, retrieval and orchestration
Embeddings	Hugging Face sentence-embedding model	768-D semantic vectors
Vector database	FAISS / ChromaDB	Vector search
Keyword retrieval	BM25	Exact-term matching
LLM	Ollama / Gemini / Hugging Face API	Generation
Agent	LangChain agent / custom ReAct	Multi-step tool use
Backend	FastAPI	Application API
Frontend	Streamlit	Working interface
Evaluation	Python metrics / RAGAS	RAG evaluation
Version control	Git/GitHub	Source submission

The supplied assignment specifically lists these tools as appropriate implementation choices. The final submission should record exact model names, versions, hardware and software environment actually used.

5. Results and Validation

5.1 Required Tests

- Run the 12-question retrieval set across at least three configurations and replace the illustrative table with actual metrics.
- Run at least three out-of-corpus questions and verify correct abstention.
- Test two departmental roles and verify that unauthorised content is never retrieved or displayed.

- Measure P50/P90 latency and per-query cost.
- Verify every citation against the cited document page.
- Record actual outputs for the four prompt variants.

Validation test	**Expected result**
Out-of-corpus question 1	Abstain
Out-of-corpus question 2	Abstain
Out-of-corpus question 3	Abstain
Authorised HR query	Correct evidence + page citation
Unauthorised department query	No unauthorised content
Citation verification	Claim supported by cited page

6. Analysis and Engineering Decision

Hybrid dense + BM25 retrieval is recommended because semantic retrieval handles paraphrases while BM25 handles exact policy terms, document names, form codes and compliance terminology. Reranking can improve precision but the supplied 120 ms per candidate makes large reranking sets expensive.

The agent should not be used for every query. Most employee questions are simple retrieval tasks and should use plain RAG. The agent should handle policy comparisons, calculations, multi-document workflows and actions such as ticket creation. This reduces latency and unnecessary tool use.

The latency calculation exposes a critical constraint: 250 generated tokens at 45 tokens/second require about 5.56 seconds. Therefore, the original 3-second target is impossible without changing the generation assumption. Shorter answers, faster generation, or a faster model are required.

Storage is not the main bottleneck: the calculated raw vector payload is only about 0.332 GB. Operational priorities are latency, access control, retrieval quality, model cost, document growth and evaluation.

7. Broader Considerations

7.1 Reliability & Safety

- Evidence thresholds and abstention prevent unsupported answers.
- Citation validation checks claims against source pages.

- Role-based filtering prevents unauthorised retrieval.
- Sensitive tool actions require verified evidence and logged results.
- Human escalation handles ambiguous or high-impact requests.

7.2 Sustainability & Economics

- Retrieval sends only relevant context to the LLM.
- Caching reduces repeated embedding and generation work.
- Selective reranking reduces cross-encoder computation.
- Local inference can reduce hosted API expenditure when appropriate.

7.3 Accessibility & Society

- Use clear, concise language for non-technical employees.
- Support paraphrased questions for non-native English speakers.
- Show source pages so employees can verify answers.
- Use a responsive interface with understandable abstention messages.

7.4 Ethics & Professional Responsibility

- Disclose AI-generated responses.
- Protect confidential company and employee information.
- Maintain auditable logs and controlled retention.
- Provide human escalation and monitor failure patterns.
- Use automation to support, not blindly replace, human judgement.

8. Conclusion

KnowledgeDesk should combine a grounded role-based prompt, hybrid RAG retrieval, selective reranking, a bounded tool-using agent and a multimodal ingestion pipeline. The context calculation permits at most 12 retrieved 512-token chunks under the stated prompt/history/answer budgets. The calculated monthly token cost for 20,000 full-budget queries is Rs.32,318.40, below Rs.40,000, and the raw vector payload is comfortably below 100 GB.

The main limitation is latency: at 45 tokens/second, a 250-token answer takes about 5.56 seconds. Therefore, the final implementation must shorten the normal answer, increase generation throughput, or use a faster model before claiming the 3-second target is satisfied.

9. Student Reflection

This assignment helped me understand that a reliable Generative AI system is more than an LLM prompt. Prompt engineering controls model behaviour, while RAG controls the evidence available to the model. I also learned that agents are useful for multi-step tasks, but using an agent for every query increases latency and complexity.

The multimodal part showed that scanned documents, tables and screenshots need more than normal text extraction. OCR is useful for exact text, while vision-based processing can preserve visual meaning. Combining these representations gives a more practical enterprise solution.

The quantitative analysis also showed how engineering decisions interact. Storage and token cost can remain within limits while generation speed still violates the latency requirement. With more time, I would evaluate real employee query logs, improve the embedding model for enterprise terminology, add a stronger reranker, improve multimodal retrieval and perform larger security tests.

This work maps to CO2 through prompt engineering and LLM behaviour, CO3 through RAG and retrieval evaluation, and CO4 through agents, multimodal processing and intelligent applications. It also supports SDG 9 through AI-enabled industry, innovation and infrastructure.

10. References

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9. J. Gao et al., “Retrieval-Augmented Generation for Large Language Models: A Survey,” 2023.
10. LangChain Documentation – retrieval, agents and vector stores.
11. FAISS Documentation – similarity search and vector indexing.
12. Hugging Face Documentation and model cards for sentence-embedding models.
13. FastAPI Documentation – API development framework.
14. Streamlit Documentation – intelligent application interface development.
15. RAGAS Documentation – RAG evaluation concepts.
16. CSA65 – Generative AI and Large Language Models, supplied Common Course Assignment, 2026–2027.

11. Individual Contribution

Student / Register No.	Responsibilities	Design & Development	Testing & Analysis	Report	Contribution
Jabastian R	Complete project planning and implementation	Prompt design, RAG pipeline, agent workflow, multimodal design and application integration	Test cases, retrieval/latency analysis, validation and debugging	Prepared major sections, tables, analysis and reflection	100%

12. Submission & Academic Integrity Checklist

- Replace illustrative retrieval metrics with actual measured values before submission.
- Insert screenshots of code execution, prompt outputs, retrieval results and application UI.
- Record exact model/version, hardware and software environment.
- Include the GitHub repository link containing source code.
- Verify every citation against the actual document page.
- Disclose AI-assisted writing/coding according to faculty policy.
- Do not present illustrative values as real experimental results.